

CODSOFT Python Programming Internship

Task 1: To-Do List Application

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➤ Internship Domain:

Python Programming

Project

****To-Do List Application Using Python****

Objective

The goal of this project is to build a command-line To-Do List Application using Python. This application helps users manage their tasks easily by allowing them to add, view and remove tasks from a task list.

Problem Statement

Managing daily tasks manually can be hard. Take a lot of time. A To-Do List Application provides a way to organize tasks, track activities and boost productivity.

The application should let users:

- * Add tasks
- * View existing tasks
- * Remove completed tasks
- * Exit the application safely

Tools and Technologies Used

Tool	Purpose
Python 3	Programming Language
VS Code / IDLE	Code Editor
Command Prompt / Terminal	Program Execution

System Requirements

- **Hardware Requirements**
 - * Computer/Laptop
 - * Minimum 2 GB RAM
- **Software Requirements**
 - * Python 3.x
 - * Any Code Editor (VS Code, PyCharm, IDLE)

Algorithm

- **Step 1**

Create a list called `TASK\_LIST` .

- **Step 2**

Show the menu options:

1. Insert Task
2. Remove Task
3. Display Tasks
4. Exit

- **Step 3**

Get user input.

- **Step 4**

Do the selected operation:

- * Insert task into the list.
- * Remove task from the list.
- * Display all tasks.

- **Step 5**

Repeat until the user chooses Exit.

Source Code

```
# To-Do List Application

TASK_LIST = []

def enqueue():

    element = input("Enter elements to insert:")

    TASK_LIST.append(element)
```

```
print(f"{element} is inserted into the task list.")
```

```
def dequeue():
```

```
    if not TASK_LIST:
```

```
        print("TASK LIST is empty! cannot remove elements.")
```

```
    else:
```

```
        removed = TASK_LIST.pop(0)
```

```
        print(f"{removed} has been removed from the task list.")
```

```
def display():
```

```
    if not TASK_LIST:
```

```
        print("TASK LIST is empty.")
```

```
    else:
```

```
        print("Current TASK LIST:", TASK_LIST)
```

```
while True:
```

```
    print("\n--- TO DO APPLICATION ---")
```

```
    print("1. INSERT TASK")
```

```
    print("2. Remove Task")
```

```
    print("3. Display Tasks")
```

```
    print("4. Exit")
```

```
choice = input("Enter your choice (1-4):")
```

```
if choice == '1':
```

```
    enqueue()
```

```
elif choice == '2':
```

```
    dequeue()
```

```
elif choice == '3':  
    display()  
elif choice == '4':  
    print("Exiting.... Thank you!")  
    break  
else:  
    print("Invalid choice! Please enter between 1-4.")
```

- **Explanation of the Code**

TASK_LIST

Stores all tasks entered by the user.

enqueue()

Adds a new task to the task list.

dequeue()

Removes the first task from the list.

display()

Displays all available tasks.

while Loop

Keeps the application running until the user selects Exit.

- **Sample Output**

--- TO DO APPLICATION ---

1. INSERT TASK
2. Remove Task
3. Display Tasks
4. Exit

Enter your choice (1-4): 1

Enter elements to insert:

Complete Python Project

Complete Python Project is inserted into the task list.

Display Tasks

Current TASK LIST:

['Complete Python Project']

Remove Task

Complete Python Project has been removed from the task list.

Features

- Add Tasks
- View Tasks
- Remove Tasks
- Easy to Use
- Menu Driven Program
- Fast and Lightweight

Advantages

- * Helps organize work.
- * Easy to understand and use.
- * Good for beginners to learn Python.
- * Shows how to use lists and functions.

Learning Outcomes

Through this project I learned:

- * Python Lists
- * Functions
- * Loops
- * Conditional Statements
- * User Input Handling
- * Menu Driven Programming
- * Basic Data Structure Operations

Future Enhancements

- * Save tasks to a file.
- * Add task priorities.
- * Add dates.
- * Create a graphical user interface (GUI) using Tkinter.
- * Store tasks in a database.

#Conclusion

The To-Do List Application was successfully built using Python. The application provides an effective way to manage daily tasks. It shows how to use Python lists, functions, loops and conditional statements while offering a solution, for task management.

Result

The To-Do List Application was successfully. Tested. Users can add, view and remove tasks through a menu-driven interface.